Electrical & Electronic Properties – Part 1

MO-201 Materials Science (Odd Sem 2026-27)

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Reference Textbooks

Materials Science & Engineering by V. Raghavan

Fundamentals of Materials Science & Engineering by W.D. Callister

Electrical & Electronic Properties

- The electrons in the outermost orbitals of the atoms of a solid determine its electrical properties

The Resistivity Range

- Electrical resistivity (or conductivity) varies over 25 orders of magnitude

The Resistivity of Materials (ohm m)

10^{-9}	10^{-7}	10^{-5}	10^{-3}	10^{-1}	10^{1}	10^{3}
Ag Cu Al Au	Ni Fe	Sb Bi Graphite	Ge (doped)	Ge (pure)		Si (pure)
← **Metals** →			← **Semiconductors** →			

10^{5}	10^{7}	10^{9}	10^{11}	10^{13}	10^{15}	10^{17}
Window glass		Bakelite	Porcelain Diamond Rubber, Nylon Polyethylene	Lucite Mica	PVC	SiO_2 (pure)
← **Insulators** →						

Electrical & Electronic Properties

- **Conductors** are metals and alloys (Au, Ag & Cu are among the best conductors of electricity, followed by Al)
- The electrical resistivity of conductors ranges from 10^{-9} to 10^{-3} ohm-m
- Electrical conductivity, being the reciprocal of resistivity, ranges from 10^{9} $ohm^{-1}m^{-1}$ to 10^{3} $ohm^{-1}m^{-1}$
- **Semiconductors** have resistivity is in the range 10^{-3} – 10^{3} ohm-m
- They form the base for a number of solid state devices
- Here, the resistivity is a strong function of small concentrations of impurities - doped Ge, with an impurity content of a few tens per million, can have a resistivity about two orders of magnitude lower than that of pure Ge
- **Insulators** have resistivity in the range of 10^{4} to beyond 10^{17} ohm-m (common examples are polyethylene, bakelite, lucite, mica, PVC, rubber and porcelain)

Electrical & Electronic Properties

- **Free Electron Theory**

- The conducting properties of a solid are not a function of the total number of electrons in the solid, as only the outermost electrons of the atoms can take part in conduction; in the free electron model, the outermost electrons of an atom are not bound to that atom, but are free to move through the whole solid

- These electrons have been variously called the **free electron cloud, the free electron gas or the Fermi gas**

- In the free electron theory, the basic assumption is that the potential field due to the ion cores is uniform throughout the solid – the free electrons have the same potential energy everywhere in the solid (this PE is a finite –ve value due to electrostatic attraction between a free electron & the ion cores)

- As we are interested only in energy differences, we can assume this constant potential to be zero; then the only energy that we have to consider is the KE

- This kinetic energy is substantially lower than that of the bound electrons in an isolated atom, as the field of motion for the free electron is considerably enlarged in the solid as compared to the field around an isolated atom

Electrical & Electronic Properties

➢ Free Electron Theory

➢ Electrons have both particle-like and wave-like characteristics; the de Broglie wavelength of an electron λ is related to its momentum *mv* as

$$\lambda = \frac{h}{mv} \qquad \text{.....(1)}$$

where *h* is Planck's constant, *m* is the mass of the free electron and *v* is its velocity

➢ The wavelength is inversely related to the magnitude of the wave number vector **k**

$$k = \frac{2\pi}{\lambda} \qquad \text{.....(2)}$$

➢ As the velocity of the free electrons is much smaller than that of light, we can ignore relativistic effects and use the classical relation for kinetic energy E

$$E = \frac{1}{2} mv^2 \qquad \text{.....(3)}$$

➢ Substituting first two equations above into the third, we can write $E = \frac{h^2 k^2}{8\pi^2 m}$

.....(4)

Electrical & Electronic Properties

- **Free Electron Theory**

- The kinetic energy E increases as the square of the wave number (parabolic relation)

- As λ is inversely related to k, the electron with the largest de Broglie wavelength will have the lowest kinetic energy

- With the enlarged field of motion in the solid, the electrons can have larger wavelengths and hence lower kinetic energies

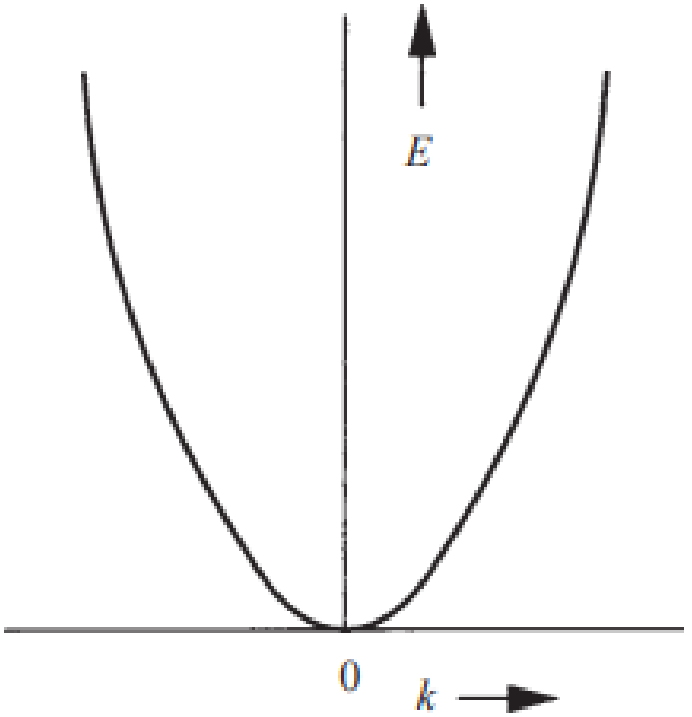


Fig. 1 The parabolic relationship between the kinetic energy E of a free electron and its wave number k

Electrical & Electronic Properties

➢ Free Electron Theory

- Even though the variation in the kinetic energy E is shown to be continuous in previous Fig., these are in fact very closely spaced discrete energy levels, as the quantum restrictions arising from the Pauli exclusion principle apply to the free electrons as well

- Let us consider a solid, in which the electron motion is unidirectional. Let L be the length of this axis of motion. The longest wavelength that is permissible for the free electron is equal to twice this length as shown in Fig. 2, as the amplitude of the wave should be zero at both ends of this length. The next smaller wavelength that is permissible is equal to L, the third one is equal to (2/3)L, and so on

- The corresponding values of the wave number k are π/L, 2π/L, … (from Eqn. (2))

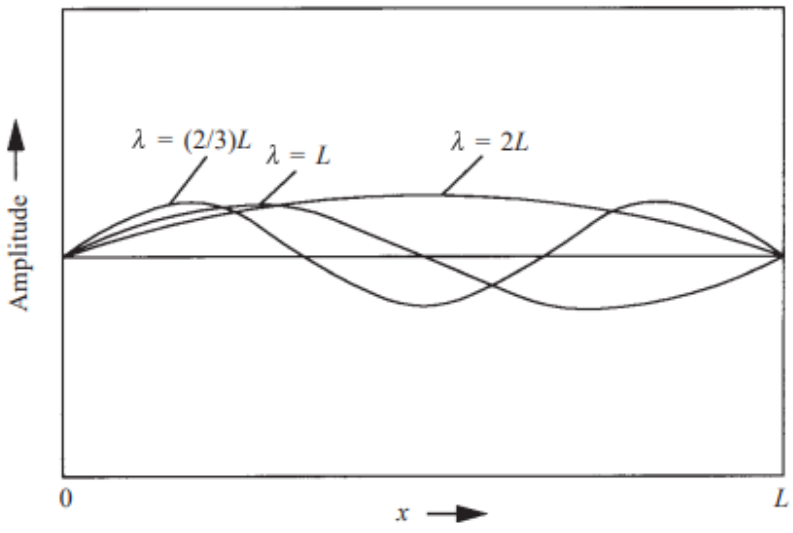


Fig. 2 The de Broglie wavelengths of the first few electrons moving along x

Electrical & Electronic Properties

- **Free Electron Theory**

- In a solid that is electrically neutral, a free electron having a certain speed in one direction will always be associated with another electron having the same speed but moving in the opposite direction

- In other words, the wave number k takes on both positive and negative values

- Thus, for the first two states, we have k equal to ± π/L and ±2π/L

- In general, k = ±nπ/L, where the quantum number n takes on successively increasing integer values. Substituting k = ±nπ/L in Eq. (4), we obtain

$$E = \frac{h^2 n^2}{8mL^2} \qquad \text{.....(5)}$$

- Equation (5) can be rewritten for three dimensional motion as

$$E = h^2 \left(\frac{n_x^2 + n_y^2 + n_z^2}{8mL^2} \right) \qquad \text{.....(6)}$$

Electrical & Electronic Properties

- **Free Electron Theory**

- Each distinct combination of n_x, n_y and n_z corresponds to a quantum state, where two electrons of opposite spins can reside

- Several combinations of n_x, n_y and n_z can result in the same value of E

- Quantum states with the same energy are said to be degenerate

- The successive energy levels calculated from Eq. (6) are so close to one another that E can be considered to be varying continuously, as illustrated for unidirectional motion in Fig. 1

- The wave forms for the free electrons starting from the lowest energy level are sketched in Fig. 2 for unidirectional motion

- The probability of finding an electron at any point along the length is proportional to the square of the amplitude of the wave at that point

- This means that, for the free electron with $\lambda = 2L$, the probability of finding it at the midpoint of the length is a maximum

Electrical & Electronic Properties

- **Free Electron Theory**

- This, however, is not an acceptable result, as the probability of finding the free electron must be the same anywhere within the solid

- To overcome this difficulty, the waves are considered to be travelling waves with a constant velocity, so that the time averaged probability of finding an electron is constant throughout the solid

- In keeping with the minimum energy criterion and the Pauli exclusion principle, the free electrons occupy successive quantum states of increasing kinetic energy

- The energy corresponding to the highest filled level at 0 K is called **Fermi energy E_F**

- At 0 K, the free electrons occupy all the levels up to the Fermi level, leaving all those above it empty

- At T > 0 K, due to thermal excitation, there is a finite probability of some of the electrons from below the Fermi level moving to levels above E_F

- This probability is given by the **Fermi–Dirac statistics**, which takes into account the quantum restrictions due to the Pauli exclusion principle

Electrical & Electronic Properties

➢ Free Electron Theory

- The probability of occupation P(E) of an energy level E by an electron is given by

$$P(E) = \frac{1}{1 + \exp\left[(E - E_F)/kT\right]}$$

- At 0 K, P(E) constant at unity with ↑ E up to E_F, where it falls abruptly to zero
- At $T_1 > 0$ K, some electrons just below E_F are thermally excited to higher levels just above E_F
- At a still higher temperature, more electrons leave the lower energy levels and occupy higher levels
- The cross-over point, where P(E) is 0.5 for different temperatures occurs at the same value of the energy level which is E_F

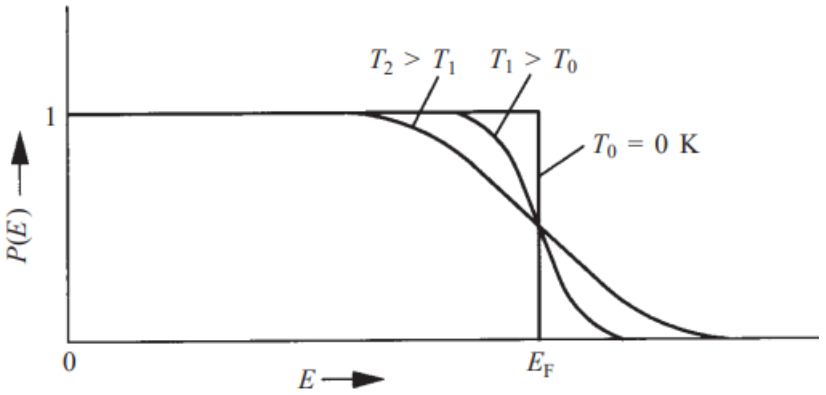


Fig. 3 The Fermi–Dirac distribution of free electrons at different temperatures

Fermi level can be defined as that level which has a 50% probability of occupation by an electron at any T

Electrical & Electronic Properties

➢ Conduction by Free Electrons

- In electrically neutral solid, for every electron moving with a certain speed in a direction, there is another electron moving with the same speed in the opposite direction
- This equal and opposite velocity distribution in a neutral solid can be biased by an externally applied electric field to yield a net velocity in one direction
- With this biasing, the solid conducts electricity
- Under an applied field, the E–k relationship of Fig. 1 gets modified to the distribution shown in Fig. 4
- Such redistribution is possible, only when empty electron states are available immediately above E_F; this availability is a basic characteristic of conductors, as opposed to semiconductors and insulators

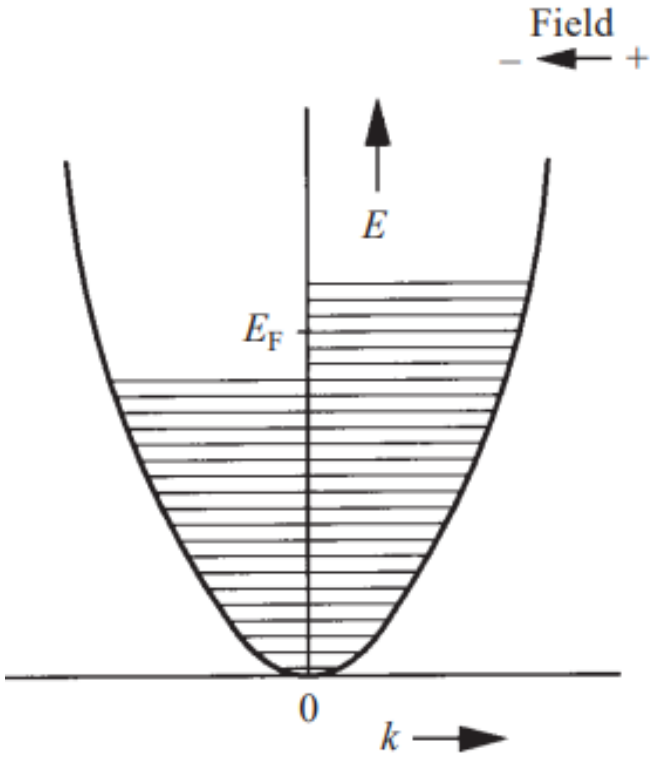


Fig. 4 Electrons moving towards the positive end of the applied field acquire extra velocity, while those moving in the opposite direction lose some velocity

Electrical & Electronic Properties

- **Conduction by Free Electrons**

- The electrons that are accelerated towards the positive end of the field do not continue to increase their velocity indefinitely

- They collide with obstacles on their way and depending on the time interval between two successive collisions, the electrons acquire an average increment of velocity called **drift velocity**, all of which they lose during a collision, as illustrated in Fig. 5

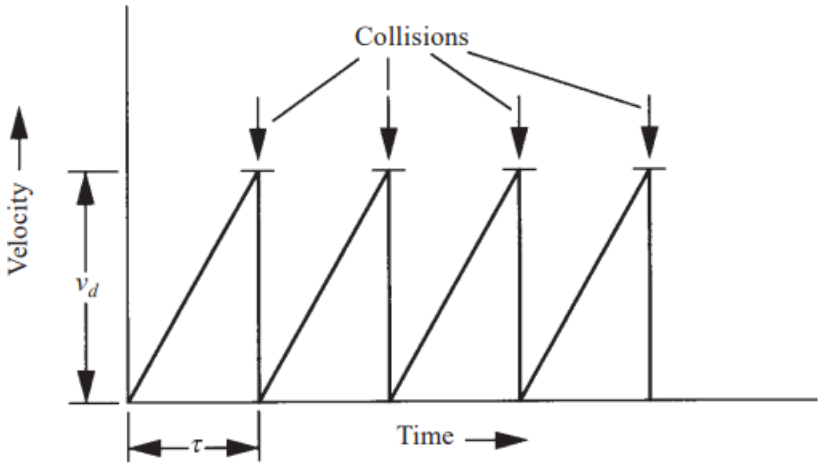


Fig. 5 The extra velocity acquired by an electron due to an applied field is lost on collision with an impurity, imperfection or phonon

- **Conductivity** is given by

$$\sigma = \frac{ne^2\tau}{m}$$

where n = no. of free electrons of charge e, τ = avg. collision time, m = mass of electron

Electrical & Electronic Properties

- **Conduction by Free Electrons**

- The mean free path l of an electron is the mean distance it travels between successive collisions

- For an ideal crystal with no impurities and imperfections, the mean free path at 0 K is infinite

- That is, there are no collisions and the electrical conductivity is ideally infinite

- Solute atoms provide effective scattering centers for the electrons

- Their introduction into the crystal results in collisions, decreasing the mean free path and the conductivity

- Similarly, other point imperfections, dislocations and grain boundaries also increase the scattering and decrease the conductivity

- Among these scattering centers, impurities dissolved in the crystal lattice as solutes are more effective than the others

Electrical & Electronic Properties

- **Superconductors**

- The resistivity of pure Ag ↓ gradually to a low but measurable value of ~10^{-11} ohm m at 0 K; Ag is not a superconductor

- The resistivity of Sn falls abruptly at about 4 K to a value below the measurable limit of 10^{-20} ohm m; the temperature at which this abrupt fall occurs is called the **superconducting transition temperature T_c**

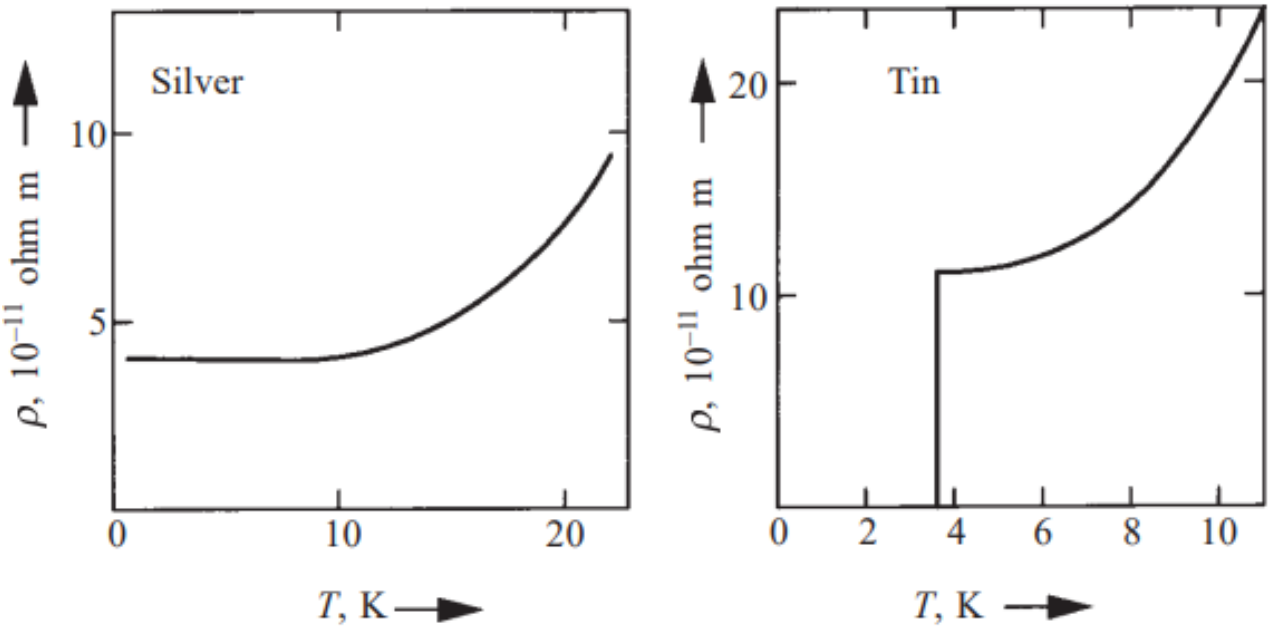


Fig. 6 The electrical resistivity ρ of (a) pure silver, and (b) tin, as a function of temperature near 0 K

Electrical & Electronic Properties

➢ Superconductors

- The first application that was conceived of the superconducting effect was to produce a large permanent magnetic field by inducing a perpetual current in a superconductor
- However, at temperatures below T_c, as the magnetic field strength reaches a critical value H_c, the superconductivity disappears; At $T = T_c$, $H_c = 0$
- H_c increases as the temperature decreases below T_c; similarly, critical current density J_c also increases as T decreases below T_c

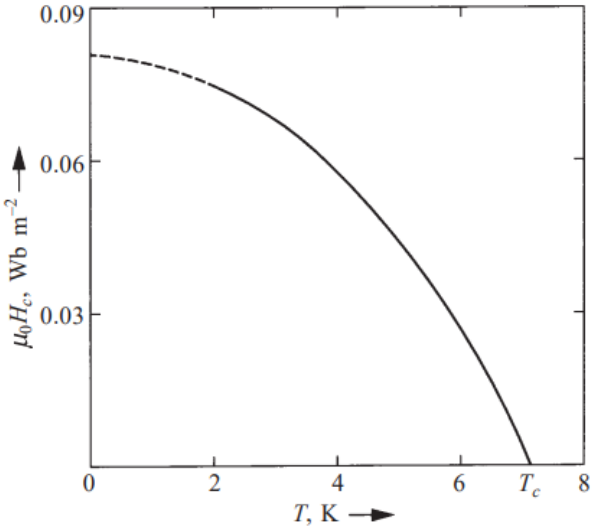


Fig. 7 The critical magnetic field H_c as a function of T for the superconducting-to-normal transition. $H_c = 0$ at T_c

In other words, the maximum current that a superconductor carries at a given T below Tc is limited by the magnetic field it produces at the surface of the superconductor

Electrical & Electronic Properties

- **Superconductors**

- In the superconducting state, the flux lines of a magnetic field are ejected out of the superconductor, as illustrated in Fig. 8; this effect is known as the **Meissner effect**

- A superconductor is a perfect diamagnetic material, with the magnetic susceptibility $\chi = -1$

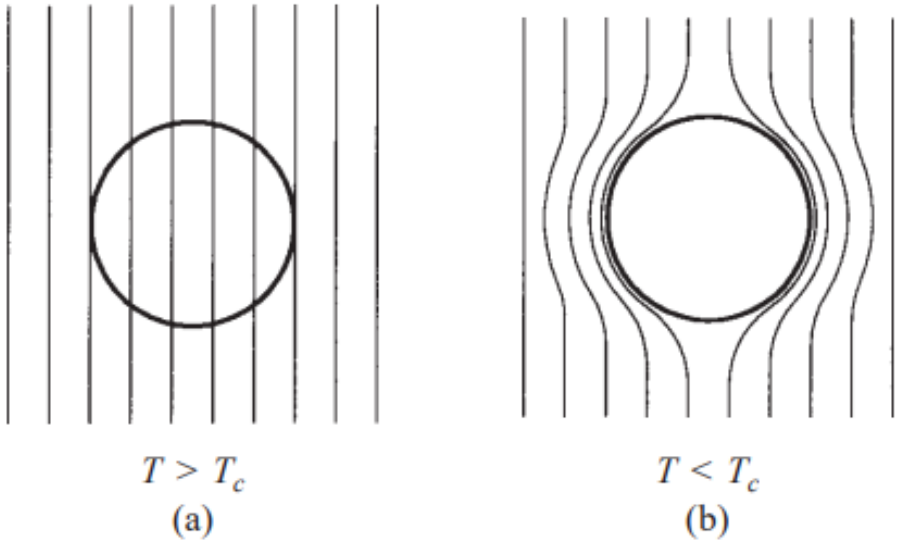


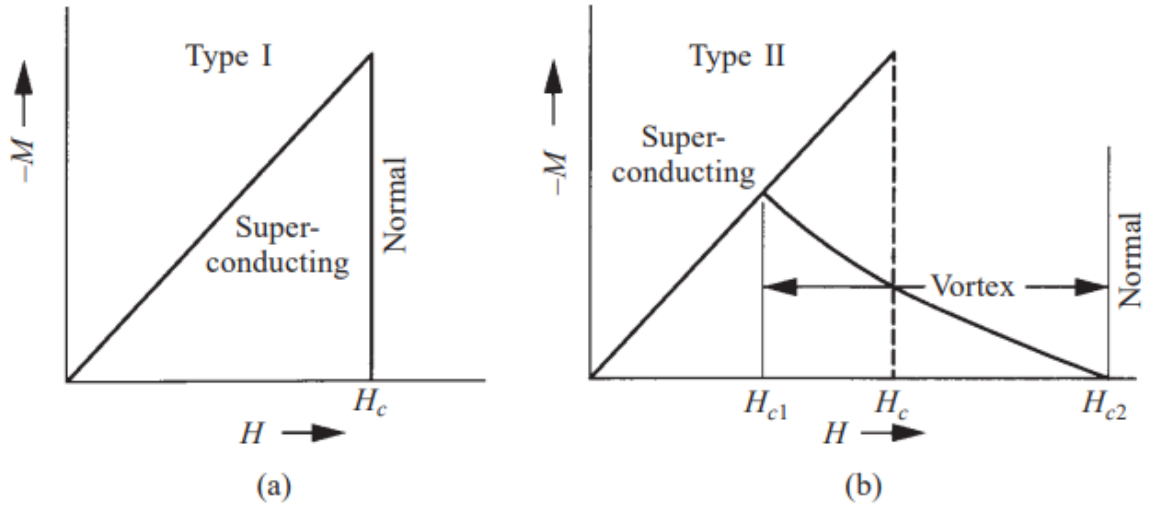
Fig. 8 The flux lines of an applied magnetic field are ejected out of the solid on crossing to the superconducting state

Electrical & Electronic Properties

- **Type I and Type II Superconductors**

- **Type I or the ideal superconductor** when placed in a magnetic field repels the flux lines totally, till the magnetic field attains the critical value H_c; the magnetization M is equal to –H up to T_c, where it drops to zero

- **Type II or hard superconductors** are those in which the ideal behaviour is seen up to a lower critical field H_{c1}, beyond which the magnetization gradually changes and attains zero at an upper critical field designated H_{c2}; Meissner effect is incomplete in the region between H_{c1} and H_{c2}; this region is known as the **vortex region**



Type II superconductors can carry high current densities, hence of practical importance

Fig. 9 The magnetization M versus the critical magnetic field H_c for (a) Type I and (b) Type II superconductors

Electrical & Electronic Properties

- **Type I and Type II Superconductors**

- Heavily cold-worked and recovery annealed material have cell walls of high dislocation density and this microstructure effectively pins the magnetic flux lines and makes their penetration difficult

- Grain boundaries also exert a pinning action; an extremely fine grain size is effective in increasing H_{c2}

- Similarly, the dispersion of very fine precipitates in the matrix with the interparticle spacing of about 300 Å results in optimum flux pinning

- The critical current density J_c also increases, as H_{c2} increases

Electrical & Electronic Properties

- **Semiconductors**

- A number of solid state devices have come into use in the last several decades, using semiconductor crystals

- The field of electronics was revolutionized since the discovery of the transistor in 1948; fragile vacuum tubes largely replaced by solid state diodes & triodes

- **The Energy Gap in Solids**

- In the free electron theory, the potential field in which the electrons are moving within the crystal is assumed to be constant; in reality, this is a gross approximation

- The potential energy of an electron will be undoubtedly a function of its position with respect to the ion cores and the other electrons in the system; e.g. its potential energy will be less if it is close to a (positive) ion core and more than the average when it is farthest away from neighbouring ion cores

- This fact combined with the diffracting properties of electrons has an important effect on the electron energy calculations, especially at certain critical values of the de Broglie wavelength of the electrons

Electrical & Electronic Properties

➢ The Energy Gap in Solids

- Consider motion of electrons along one axis (same as before); electrons moving to the right or to the left are scattered by the atoms in the path

- This scattering is largely coherent and is being reinforced in the direction of motion of the electrons; if, however, the spacing d between neighbouring atoms along the axis is such that the Bragg equation is exactly satisfied, the electrons are totally reflected

- For the unidirectional motion under consideration, the atoms along the axis can be visualized as successive ‘planes’ perpendicular to the direction of electron motion so that $\sin \theta = 1$. Then, from Bragg equation, the wavelength λ of the electron takes on critical values of 2d, d, 2d/3, and so on, for successively increasing order of reflection, denoted by n

- Expressed in terms of the wave number k ($k = 2\pi/\lambda$) we get the critical conditions at $k = \pm\pi/d, \pm 2\pi/d, \pm 3\pi/d, \ldots$ the electrons are reflected back and forth repeatedly, when they have the critical wave number

- We can say that the net velocity of the electrons in a given direction is zero, due to repeated reflection

Electrical & Electronic Properties

- **The Energy Gap in Solids**

- The wave form of the electrons under such critical conditions comes out to be two standing waves rather than the travelling wave we visualized for the free electron

- The travelling wave gives a constant, time-averaged density of electrons throughout the solid, while the standing wave, gives a periodic variation in the amplitude and in the electron probability density in the crystal

- Two possible combinations (the sum and the difference) of the travelling waves give respectively a cosine wave and a sine wave

- These waves have the maximum amplitude at fixed points in space

- The probability of finding an electron at a point is proportional to the square of the amplitude of the wave at that point

- The electron is most likely to be found at the crest of the probability density curve; in other words, the electron is localized here, in contrast to the free electron, which can be found anywhere within the crystal

Electrical & Electronic Properties

➢ The Energy Gap in Solids

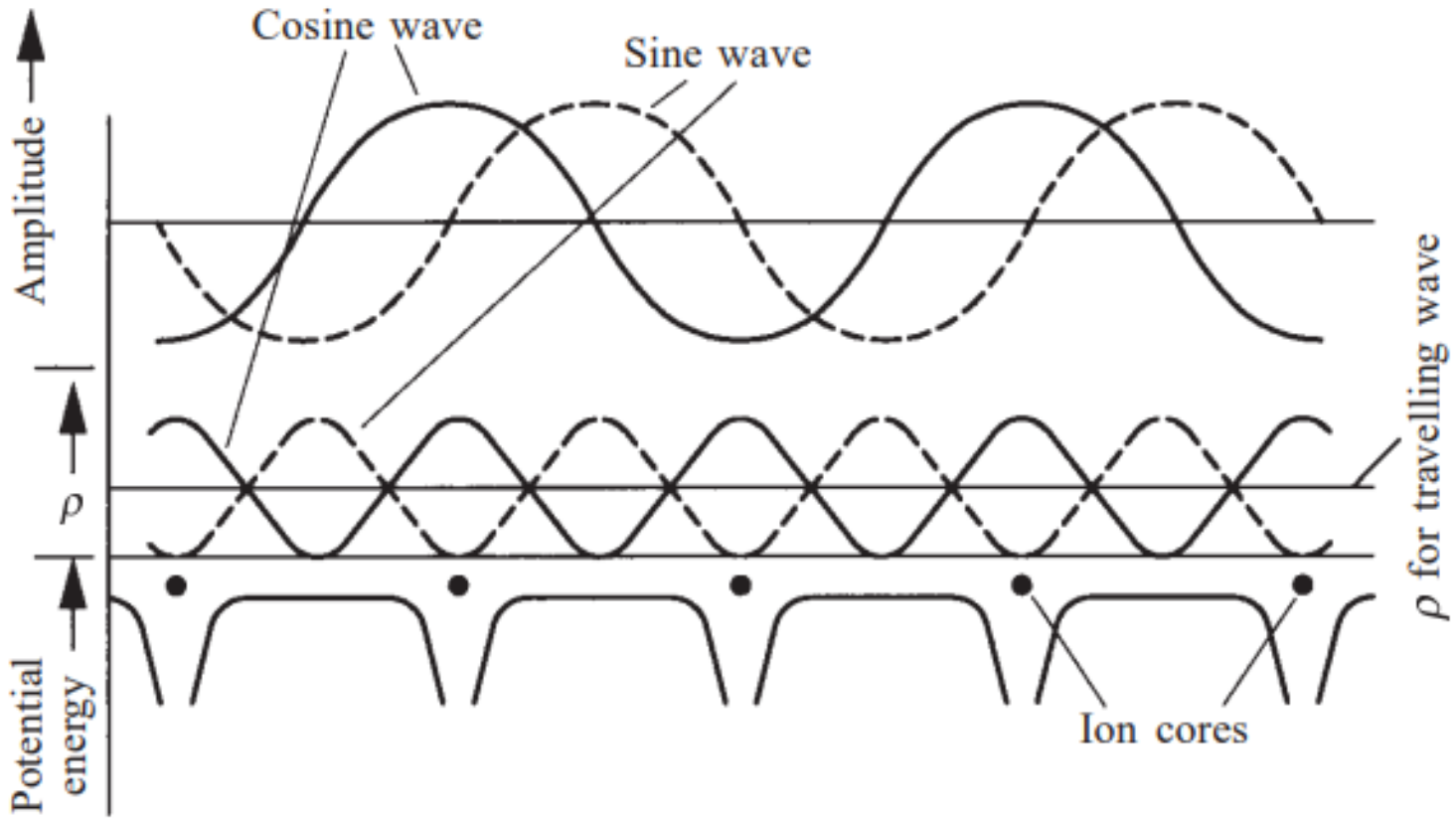


Fig. 10 The wave form of electrons are two standing waves when the Bragg law is satisfied. The electron probability density ρ is shown for the two waves. At the bottom, the potential energy variation as a function of electron position is sketched.

Electrical & Electronic Properties

- **The Energy Gap in Solids**

- When the electrons are localized, the potential energy variations due to location have also to be taken into account

- When there is no critical condition and the electrons are not reflected, they have the average potential energy, which can be taken to be zero

- At a critical condition, however, the electron is described by a standing wave and is therefore localized

- When the wave form is a standing cosine wave, the electron is close to an ion core; therefore, it has a lower potential energy than the average

- When the wave form is a standing sine wave, the electron is farthest away from the ion cores and correspondingly has a potential energy greater than the average

Electrical & Electronic Properties

➢ The Energy Gap in Solids

- These deviations from the average potential energy appear as a break in the E–k curve
- The break occurring at a critical value of k gives rise to an energy gap
- Two groups of very closely spaced energy levels called energy bands are separated by an energy gap at the critical value of k
- The magnitude of this energy gap is the difference in the potential energy of the two electron locations

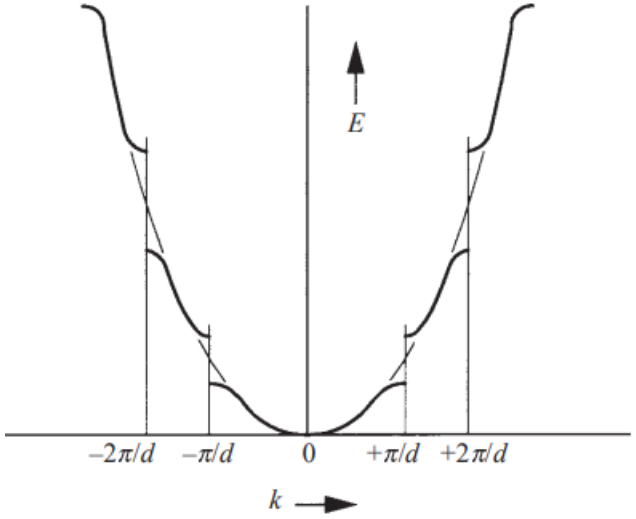


Fig. 11 E–k plots show a break in the curve at the critical values of the wave number k

- In the 3D crystal, the electrons can move in any direction not necessarily normal to the reflecting planes – the Bragg angle θ can have any value from 0 to 90°
- Critical values of k will then correspond to $\pm n\pi/d\sin\theta$ and not $\pm n\pi/d$, which we used for the 1D case

Electrical & Electronic Properties

➢ The Energy Gap in Solids

- Depending on the value of θ, reflection will occur at different values of k, even for the same set of reflecting planes
- The E–k curves for electron motion along [100] and [110] directions are shown in Fig. 12 for first order reflection from (100) planes
- The gap in the E–k curve occurs at $k = \pi/d$ for [100] motion and at $k = \sqrt{2}\pi/d$ for [110] motion
- The effective gap is between the bottom line of the gap in the [110] direction and the top line of the gap in the [100] direction
- This effective gap is called the **forbidden gap**, because the electrons cannot take those values of energy that lie in the forbidden gap, regardless of their direction of motion

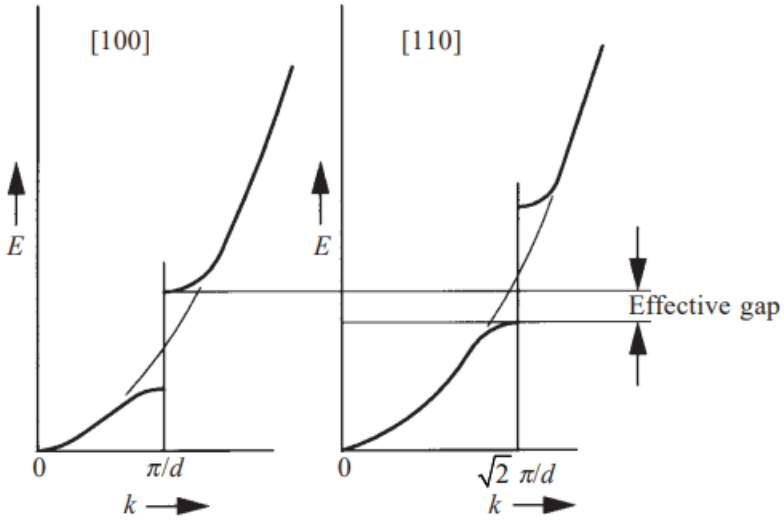


Fig. 12 The effective forbidden gap refers to those energy values which an electron cannot take, whatever be the direction of motion in the crystal